When the Target Is Wrong

Reframing Plant Radiation Dosimetry as Cross-Study Exposure Triage

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Predicting absorbed radiation dose from gene expression is an attractive framing, but it is only answerable if the underlying experiments actually span the dose axis. Auditing a six-study collection of *Arabidopsis thaliana* RNA-seq profiles (158 samples, 4,002 genes, NASA OSDR), we find that 138 of 158 samples sit at exactly 0 or 100 Gy and only 20 represent intermediate doses—a design that supports exposed-versus-control triage but not a calibration curve. We therefore change the target and report both outcomes. Under leave-one-study-out validation, in which an entire study is withheld and all adaptive operations including gene selection are fitted inside the training folds, a 50/100-gene stability ensemble reaches 84.2% pooled accuracy, 82.4% balanced accuracy, 0.883 pooled ROC-AUC, 0.656 Matthews correlation, 89.0% sensitivity, and 75.9% specificity. The same protocol applied to exact-dose regression yields 33.6 Gy mean absolute error and Spearman $\rho = 0.523$: the negative result that motivated the reframing. Because the independent unit is a study and not a sample, we bootstrap over the six study-level scores rather than over the 158 samples, giving a macro balanced accuracy of 0.849 with a 95% interval of $[0.757, 0.938]$—broad, and honestly so. Three canonical DNA-damage-response genes are selected in every outer fold, which makes the classifier biologically plausible without establishing that its markers are specific to radiation rather than to stress in general.

CCS Concepts: • **Applied computing: Bioinformatics** • **Applied computing: Life and medical sciences** • **Computing methodologies: Supervised learning by classification**

Additional Key Words and Phrases: Arabidopsis, domain generalization, gene expression, leave-one-study-out validation, radiation biology, study design, triage

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1 INTRODUCTION

A predictive question is only as good as the experimental design behind it. Asking a model to estimate absorbed radiation dose from a transcriptome presupposes that the available data samples the dose axis densely enough for a dose–response relationship to be identifiable. When it does not, a regression can still be fitted, a correlation can still be reported, and the resulting number can look like partial success while in fact measuring something closer to group separation.

This paper is a worked example of catching that failure and responding to it. Working with six NASA Open Science Data Repository studies of irradiated *Arabidopsis thaliana* [12], we audit the target before modelling it, find that the dose axis is effectively binary, and reframe the task as the operational question the data can support: *given an expression profile from a study never seen during training, was this plant irradiated?*

1.1 Contributions

(1) A target-design audit that rejects the originally posed regression on identifiability grounds, with the failed regression retained and reported rather than discarded.
(2) A leakage-controlled cross-study benchmark in which the held-out unit is an entire study and all feature selection, scaling, and fitting occur inside the training folds, comparing a literature panel against linear, kernel, tree, and ensemble models.
(3) Uncertainty quantified at the correct level of analysis. With six independent studies, we bootstrap study-level scores rather than samples, which widens the intervals substantially and reflects what the design actually supports.
(4) A stability analysis of repeatedly selected genes, with an explicit statement of why concordance with the DNA-damage-response literature is plausibility evidence and not biomarker validation.

2 DATA AND DESIGN AUDIT

2.1 Cohort

The harmonized matrix contains 158 samples × 4,002 genes drawn from six studies (OSD-498, 502, 508, 510, 658, 782), with a manifest recording absorbed dose, radiation quality (gamma or heavy ion), genotype, and tissue. Of the 158 samples, 58 are unirradiated controls and 100 are exposed.

2.2 Why exact dose is not identifiable here

Figure 1(a) shows the distribution of absorbed dose. Four of the six studies contain only 0 and 100 Gy. Pooling everything, 138 of 158 samples (87%) sit at one of those two extremes; 10, 40, and 80 Gy together contribute 20 samples, with only four samples at each of 40 and 80 Gy.

A regression trained on this support cannot distinguish a genuine dose–response gradient from a two-group contrast. Reporting Spearman $\rho \approx 0.52$ for such a model is technically correct and substantively misleading: with 87% of the mass at two points, a rank correlation is dominated by the model's ability to separate those two points, not by any calibration in between. We therefore change the endpoint to

$$y_i = \mathbb{1}\left[\text{dose}_i > 0\ \text{Gy}\right], \tag{1}$$

and revisit the regression in Section 4.6 as a reported negative result.

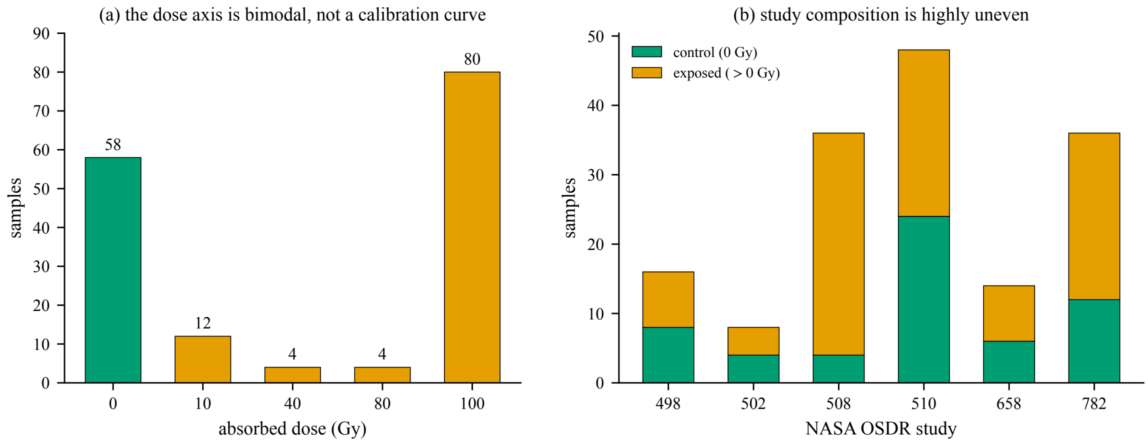

Fig. 1. (a) The absorbed-dose distribution is bimodal: 138 of 158 samples lie at 0 or 100 Gy, and the intermediate levels carry 12, 4, and 4 samples respectively. (b) Study composition is highly uneven; OSD-508 is 88.9% exposed while OSD-510 is balanced, so any protocol that ignores study membership will confound exposure with batch.

2.3 Preprocessing provenance

The distributed matrix is log-transformed, per-study standardized, and restricted to the 4,002 most variable genes. Per-study standardization suppresses laboratory-scale batch effects [8], but it is transductive: it uses the distribution of an entire study, including its test samples, before any label is predicted. Our benchmark is therefore conditional on this supplied representation. A deployable pipeline would require a reference-based normalization that can transform a single incoming sample. Concretely, a reference panel must be frozen using training studies; library-size or variance-stabilising transforms, per-gene centres and scales, and quality-control limits must then be learned from that panel and applied unchanged to each incoming sample. External spike-ins or validated housekeeping genes could anchor the transform, while reference samples processed with every laboratory batch would monitor drift. No cohort statistics from the incoming study may enter prediction. Developing and prospectively validating that single-sample transform is the principal gap between this benchmark and a usable assay.

3 EVALUATION PROTOCOL

3.1 Leave-one-study-out

Randomly splitting samples would place the same laboratory, genotype, growth protocol, and sequencing batch on both sides of the split, estimating interpolation within known studies rather than transfer to a new one [14]. We instead hold out one complete study at a time:

$$\hat{p}^{(s)} = f_{\mathcal{D}\setminus s}(\mathbf{x}), \qquad \mathbf{x} \in s, \quad s \in \mathcal{S}, \ |\mathcal{S}| = 6. \tag{2}$$

Every adaptive operation—the ANOVA F-test gene selector, the scaler, and the classifier—is refitted on $\mathcal{D} \setminus s$ alone. Selecting genes on the full matrix before cross-validation is the classic inflation mechanism in expression studies [1], and Equation (2) is specified precisely to exclude it. The decision threshold is fixed at 0.5 throughout and is never tuned.

Table 1. Leave-one-study-out performance, six folds, threshold fixed at 0.5. All adaptive operations are fitted inside the training studies.

Model	Acc.	Bal. acc.	ROC-AUC	MCC	Sens.	Spec.
Stability ensemble	**0.842**	**0.824**	0.883	**0.656**	0.890	0.759
Top-100 ExtraTrees	0.823	0.806	**0.886**	0.616	0.870	0.741
Top-50 logistic	0.791	0.799	0.859	0.579	0.770	**0.828**
Top-100 RBF-SVM	0.804	0.783	0.878	0.574	0.860	0.707
DDR-7 logistic	0.741	0.770	0.741	0.521	0.660	0.879
Always exposed	0.633	0.500	0.500	0.000	1.000	0.000

3.2 Metrics

Because 63.3% of samples are exposed, accuracy alone is inadequate: the trivial always-exposed rule scores 0.633. We therefore report balanced accuracy, ROC-AUC [6], average precision, sensitivity, specificity, and the Matthews correlation coefficient

$$\mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN} - \mathrm{FP}\cdot\mathrm{FN}}{\sqrt{(\mathrm{TP+FP})(\mathrm{TP+FN})(\mathrm{TN+FP})(\mathrm{TN+FN})}}, \tag{3}$$

which uses all four confusion-matrix cells and is the appropriate single summary under class imbalance [2].

3.3 Model ladder

Five models plus a baseline are compared, in increasing flexibility: (i) the always-exposed rule; (ii) DDR-7, logistic regression on seven literature DNA-damage-response genes; (iii) top-50 logistic regression with fold-local selection; (iv) a top-100 RBF-SVM; (v) top-100 extremely randomized trees [7]; and (vi) a stability ensemble averaging tree probabilities from the top-50 and top-100 training-fold panels. The ensemble is motivated by stability selection [9]: averaging over two panel sizes removes the arbitrary choice of a single k.

4 RESULTS

4.1 Cross-study benchmark

Table 1 and Figure 2 give the leave-one-study-out results.

Three comparisons matter. First, the always-exposed rule makes the case for balanced metrics concrete: 0.633 accuracy, 0.500 balanced accuracy, zero specificity. Second, the seven-gene literature panel is genuinely informative (MCC = 0.521) and has the second-highest specificity in the table, but its ROC-AUC of 0.741 is well below every fold-local multigene model—the canonical DDR genes carry real signal but not all of it. Third, the stability ensemble gives the best thresholded performance (MCC = 0.656), though its pooled AUC (0.883) is statistically indistinguishable from the single top-100 tree model (0.886). The ensemble's advantage is in calibration at the fixed threshold, not in ranking.

4.2 Error structure

Figure 3 decomposes the ensemble's errors. The pooled confusion matrix is TP = 89, FN = 11, TN = 44, FP = 14: sensitivity 89.0% against specificity 75.9%. The asymmetry is consistent throughout the model ladder and is the operationally important weakness—roughly one unirradiated control in four is flagged as exposed, which for a screening instrument means the positive calls require confirmation.

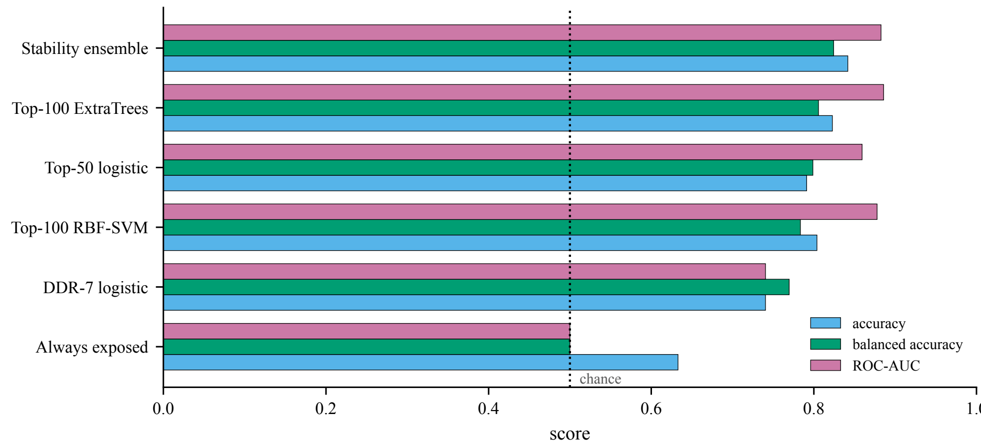

Fig. 2. The model ladder. The always-exposed rule attains 0.633 accuracy with 0.500 balanced accuracy and zero specificity, which is exactly why accuracy is not the primary endpoint. Fold-local multigene models improve substantially on the literature panel.

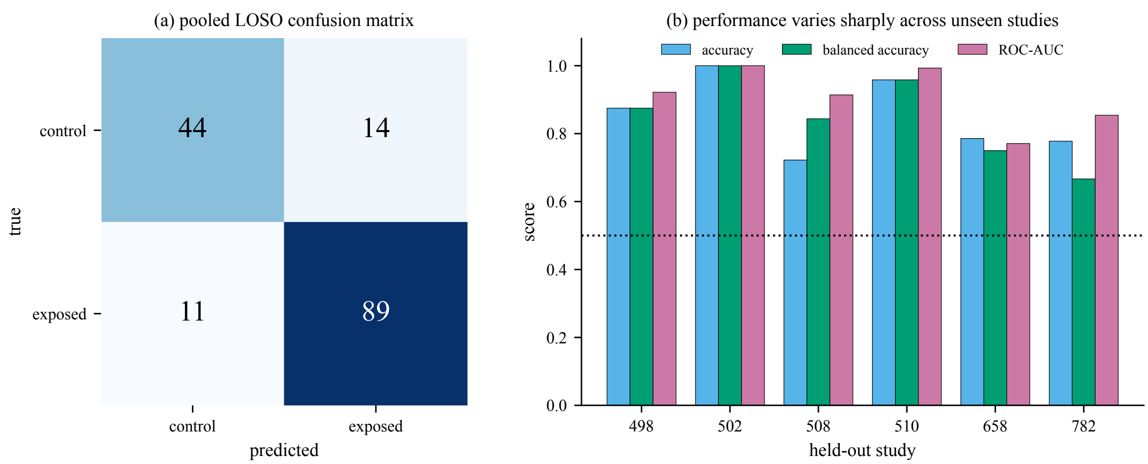

Fig. 3. (a) Pooled confusion matrix for the stability ensemble. Specificity (75.9%) lags sensitivity (89.0%). (b) Per-study performance. OSD-502 is perfectly classified and OSD-510 nearly so, while OSD-782 falls to 66.7% balanced accuracy despite an AUC of 0.854—a threshold problem, not a ranking failure.

Per-study performance ranges widely: OSD-502 is perfectly separated (AUC = 1.000, n = 8), OSD-510 nearly so (0.993, n = 48), while OSD-658 falls to 0.771 and OSD-782 to 0.667 balanced accuracy. OSD-782's case is diagnostic. Its AUC is a respectable 0.854, so the ranking is sound; what fails is the fixed 0.5 threshold on that study's score distribution. A held-out study can therefore be well-ordered and still be badly classified, which is an argument for reporting both a ranking and a thresholded metric rather than either alone.

4.3 Do intermediate doses register?

The intermediate levels are far too sparse for dose–response claims, but they test whether the binary model merely memorised the 100 Gy condition. Figure 4(a) shows the mean held-out exposure score rising with dose and remaining

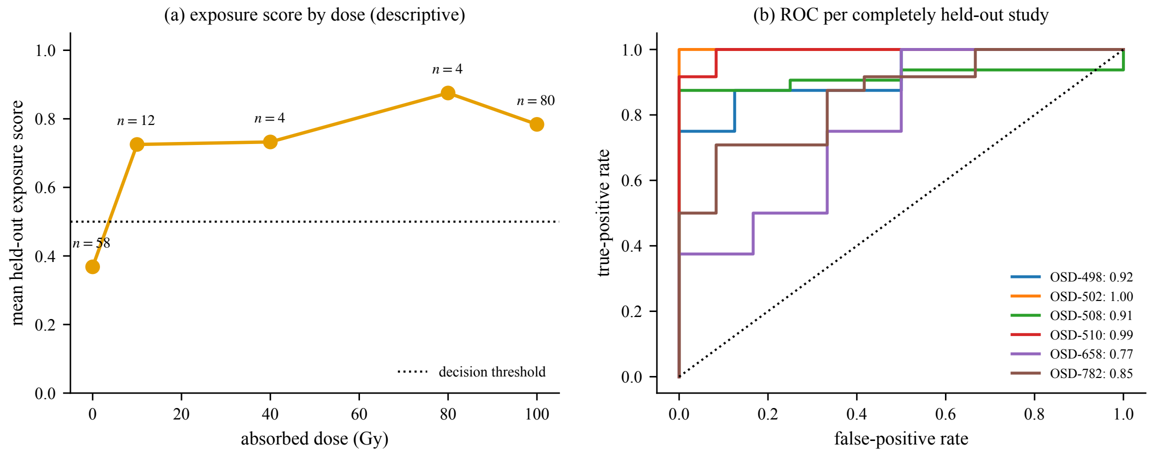

Fig. 4. (a) Mean held-out exposure score by dose. Intermediate doses score above the threshold, but with $n \leq 12$ per level these are descriptive points, not population estimates. (b) ROC curves for each completely held-out study, showing the spread that a pooled curve conceals.

Table 2. Study-level bootstrap, $B = 10{,}000$ resamples of the six held-out studies. The intervals are broad because the design supports six independent observations, not 158.

Macro-averaged metric	Estimate	95% interval
Accuracy	0.853	[0.777, 0.937]
Balanced accuracy	0.849	[0.757, 0.938]
ROC-AUC	0.909	[0.846, 0.970]

above the threshold at 10, 40, and 80 Gy. With $n = 12$, 4, and 4 respectively, these points are descriptive only. Dose is also confounded with study, radiation quality, and tissue, so neither monotonicity nor a dose–response relation can be inferred. The only supported reading is that the classifier is not purely a 100 Gy detector.

4.4 Uncertainty at the right unit of analysis

There are 158 samples but only six independent external validation studies. Samples within a study share genotype, growth conditions, harvest, and sequencing batch, so treating them as independent replicates would produce intervals that are far too narrow for the design [15]. We therefore bootstrap the six study-level scores [5]. Table 2 reports the result.

The macro balanced-accuracy interval spans 18 percentage points. That width is the honest headline of this study: the point estimate is encouraging, and the design cannot presently distinguish 0.76 from 0.94.

4.5 Which genes are selected consistently?

Repeating the fold-local selector and counting how often each gene enters the top-100 panel gives a stability profile that, unlike a single fit, cannot be driven by one study. Across all 4,002 genes, 186 enter at least one top-100 panel: 72 are selected in one fold, 19 in two, 8 in three, 16 in four, 24 in five, and 47 in all six (median frequency 3 among selected

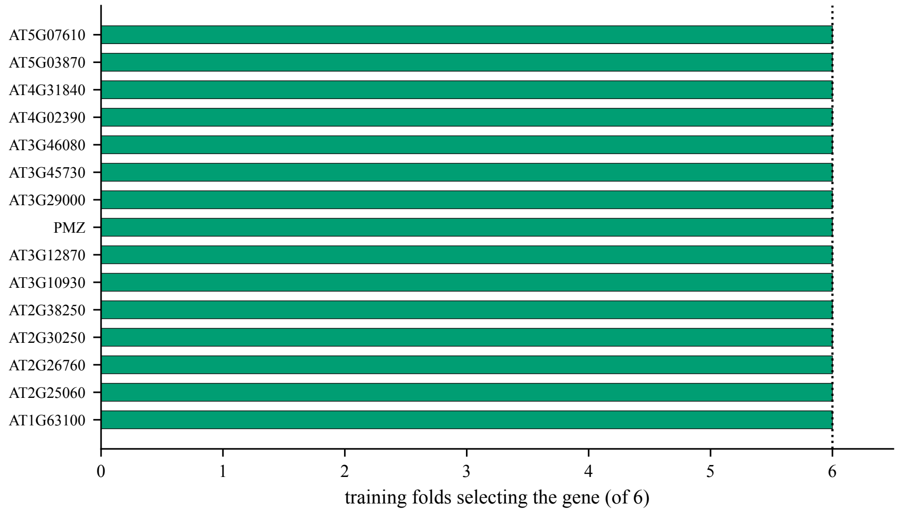

Fig. 5. Genes selected most often across the six training-fold panels. A gene reaches six only if it survives regardless of which study is held out.

genes). Figure 5 shows the most stable subset. Three canonical DNA-damage-response genes—AT4G02390, AT4G22960, and AT5G66130—are selected in all six outer folds, alongside cell-cycle, repair, and heat-shock annotated genes.

Ionising radiation can generate reactive oxygen species and DNA lesions, activating damage sensing, repair, cell-cycle arrest, and general stress programmes. The observed cell-cycle, repair, and heat-shock annotations therefore agree with known plant radiation biology [3, 10] and make the classifier biologically plausible. It does not make these genes radiation biomarkers. The same DNA-damage and oxidative programmes are induced by drought, heat, pathogens, and chemical clastogens [4, 10], and this collection contains no non-radiation stress controls. Specificity to radiation is untested and, on this data, untestable.

4.6 Exact dose remains out of reach

For continuity we repeat the original regression under the identical leave-one-study-out boundary. Gradient-boosted regression on the same matrix yields MAE = 33.6 Gy and Spearman $\rho = 0.523$ (Figure 6). Against a dose range of 0–100 Gy, a mean absolute error of a third of the range is not a usable dosimeter, and the predictions collapse toward the interior rather than tracking the diagonal.

We report this because it is the evidence for the reframing. The regression is not weak because the model is inadequate; it is weak because 87% of the design carries no information about the interior of the dose axis.

5 DISCUSSION

5.1 Supported findings

(1) Changing the question improves both the science and the measured performance: the collection supports cross-study exposure triage and does not support exact dose regression.
(2) The stability ensemble reaches 84.2% pooled accuracy, 82.4% balanced accuracy, 0.883 pooled ROC-AUC, and MCC = 0.656 while holding out one complete study at a time.

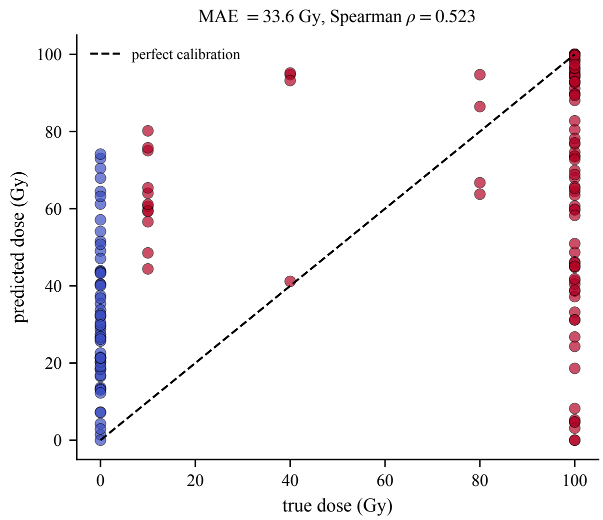

Fig. 6. Secondary task: exact-dose regression under the same leave-one-study-out protocol. Predictions cluster in the interior, the signature of a model separating two groups rather than estimating a continuous quantity.

(3) Sensitivity (89.0%) exceeds specificity (75.9%) consistently, so the model is a screening instrument whose positive calls need confirmation.
(4) Fold-local feature selection is essential; selecting genes before cross-validation would leak the held-out study.
(5) With six studies, study-level bootstrap intervals span roughly 18 percentage points of balanced accuracy.

5.2 Threats to validity

The supplied matrix is per-study standardized, which is transductive, so the benchmark is conditional on that representation and does not establish that a single new sample could be normalized and scored. Six studies is a small external sample, and OSD-508's 88.9% exposed fraction means one fold is close to degenerate. The selected genes are associated with exposure in this collection but have not been tested against non-radiation stressors. Finally, the intermediate-dose analysis rests on four samples per level and should not be read as a dose–response result.

Deployment would also cross several untested boundaries: sequencing platform and laboratory, genotype and tissue, radiation quality and time since exposure, and the prevalence-dependent calibration of the fixed 0.5 threshold. A deployable screen therefore requires a frozen single-sample normalization, laboratory quality controls, prospective calibration, an abstention rule for out-of-distribution samples, and confirmation of positive calls by an independent assay. The present classifier is a cross-study research benchmark, not a field-ready diagnostic.

5.3 The recommended next experiment

The most valuable addition is not another model or another decimal place. It is a genuinely untouched seventh study containing multiple doses, biological replicates, and non-radiation stress controls, opened only after the 50/100-gene ensemble and the 0.5 threshold have been frozen. A Fisher-z calculation requires about 29 independent samples for 80% power at two-sided $\alpha = 0.05$ to detect a correlation of 0.5. We therefore recommend at least 30 independent biological replicates across at least five prespecified doses (at least six per dose) as a minimum confirmatory curve, preferably repeated in more than one study. This is a planning lower bound, not enough by itself to establish cross-study generalisation. The experiment would also test the specificity claim that the present data cannot address.

6 MODEL CARD

Table 3. Model card [11] for the stability ensemble.

Item	Statement
Intended use	Research screening of exposed versus control *Arabidopsis* RNA-seq samples.
Evaluation population	Six NASA OSDR plant studies, 158 samples.
External unit	One completely held-out study.
Positive class	Absorbed dose greater than 0 Gy.
Primary metrics	Balanced accuracy and ROC-AUC; raw accuracy is supporting.
Decision threshold	Fixed at 0.50; never tuned.
Adaptive operations	F-test gene selection, scaling, and model fitting, all inside the training fold.
Prohibited use	Continuous-dose estimation, human dose estimation, individual safety decisions, or claims of causal biomarkers.
Known weak point	Specificity (75.9%) lags sensitivity (89.0%); only six external studies exist.

7 CONCLUSION

Auditing the target before modelling it changed this study's conclusion from a weak regression to a defensible classifier. On six NASA OSDR *Arabidopsis* studies, a stability ensemble identifies radiation exposure at 84.2% accuracy and 0.883 ROC-AUC with an entire study held out, while the exact-dose regression the design cannot support returns 33.6 Gy error. The governing uncertainty is the number of studies, not the number of samples, and the study-level intervals say so plainly.

REPRODUCIBILITY

The repository contains the harmonized matrix, sample manifest, raw-study examples, annotation tables, and environment specification in the public repository. The figure script accompanying this manuscript re-runs the entire leave-one-study-out benchmark, the study-level bootstrap, the gene stability analysis, and the secondary regression from those files in under a minute on a CPU, using scikit-learn [13]. The requirements specify Python 3.11 and minimum versions NumPy 1.24, pandas 2.0, Matplotlib 3.7, scikit-learn 1.3, SciPy 1.10, and PyArrow 12. Seeds, fold definitions, feature counts, hyperparameters, target definition, and threshold are fixed in code.

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